Accessing the speed of sound in relativistic ultracentral nucleus-nucleus collisions using the mean transverse momentum

Fernando G. Gardim, Andre V. Giannini, and Jean-Yves Ollitrault

Goal

The speed of sound of the strong interaction at high temperature can be measured using the variation of the mean transverse momentum with the particle multiplicity in ultracentral heavy-ion collisions.

Test this correspondence by running hydrodynamic simulations at zero impact parameter with several equations of state.

Discusses the differences between this ideal setup and an actual experiment.

INTRODUCTION

Primary goals of ultrarelativistic nucleus- nucleus collisions is to infer the thermodynamic properties of the quark-gluon plasma .

Thermodynamic properties are an input of hydrodynamic equations. By comparing the particle distributions measured in experiment with those calculated in hydrodynamic models, constrain these properties.

Simple proportionality relation has been observed in hydrodynamic simulations, confirming an old idea put forward by Van Hove:

$$\langle p_T \rangle \simeq 3\, T_{eff} \,,$$

where the proportionality factor 3 is purely phenomenological, while the effective temperature roughly corresponds to the average temperature at a time of the order of the nuclear radius.

INTRODUCTION

The impact parameter of an ultracentral collision is close to zero, but the multiplicity can vary by up to 10-15% due to quantum fluctuations, thus an increase of $\langle pT \rangle$ is expected as the multiplicity increase.

The temperature increase induced by an increase in density is determined by the speed of sound c_s defined by $c^2_s = d \ln T / d \ln s$, where s is the entropy density.

Since the entropy density is proportional to the particle multiplicity N_{ch}

$$c_s^2(T_{\text{eff}}) = \frac{d \ln \langle p_T \rangle}{d \ln N_{ch}}.$$

INTRODUCTION

The CMS collaboration has recently measured the speed of sound at T_{eff} = 219 ± 8 MeV with this method, and the result c^2_s = 0.241 ± 0.016 is in perfect agreement with lattice QCD calculations.

The primary goal of this paper is to assess more precisely the validity of these Eqs., which form the theoretical basis of the CMS analysis, by means of systematic hydrodynamic calculations.

METHOD

For simplicity, an ideal situation consisting of Pb+Pb collisions at zero impact parameter, b = 0, with a smooth initial density profile, and assuming exact longitudinal boost invariance is simulated. Within this simplified setup, the robustness of Eqs. against variations of the equation of state and collision energy is checked.

The default equation of state of our hydrodynamic calculation is that obtained through lattice calculations by the hotQCD collaboration. Small variations around this equation of state is studied. The priority here is to test whether we are able to capture small changes in the speed of sound through experimental data.

METHOD

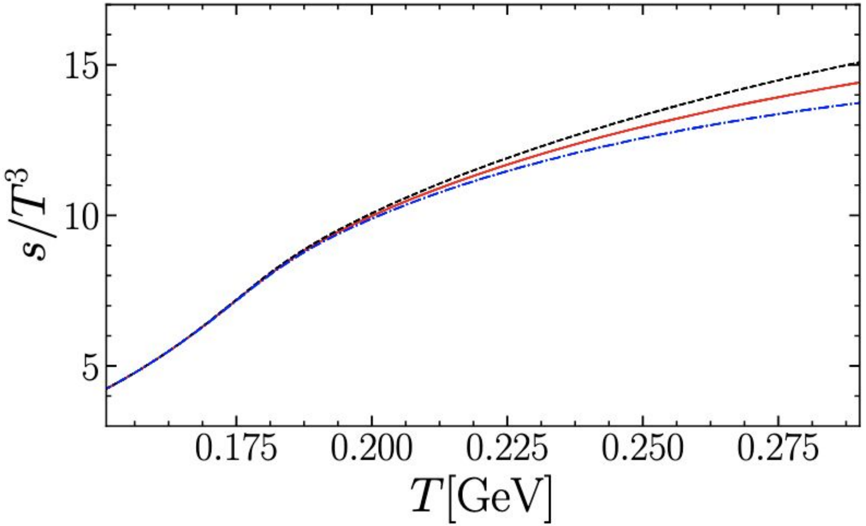
The equation of state is not modified near the freeze-out temperature $T = T_F$ at which the fluid is converted into hadrons.

Following modification of the pressure is chosen $P(T) \rightarrow P(T) + \alpha(T - T_F)^4$, T_F=151MeV.

It ensures that the modified entropy density and speed of sound vary smoothly with the temperature, and are not modified at $T = T_F$. In the limit $T \rightarrow \infty$, the ratio $P(T)/T^4$ is proportional to the number of degrees of freedom of quarks and gluons, so that α physically represents a change in the number of degrees of freedom.

METHOD

Two opposite values α = 1.54 and α = −1.54 are implemented, in addition to the default value α = 0. A positive α implies a softer equation of state, with a lower speed of sound, while a negative α implies a harder equation of state.



METHOD

Hydrodynamic calculation is that of the Duke analysis, which is calibrated to reproduce LHC data on p+Pb and Pb+Pb collisions at $\sqrt{s_{NN}}$ = 5.02 TeV.

Smooth initial density is obtained by generating 1000 events at b = 0 using the TRENTo model. After run assess precisely the effect of a variation in multiplicity, by rescaling the initial energy density profile by a multiplicative constant.

Evolved through the hydrodynamic code MUSIC after a free-streaming stage. Interactions in the subsequent hadronic phase are modeled with the UrQMD transport code.

Running the transport calculation 1000 times, which reduces the statistical errors

METHOD

For each hydrodynamic simulation, we evaluate the total energy E and entropy S of the fluid at the temperature where it is converted into hadrons, by integrating over the hypersurface $T = T_F$ for each hydrodynamic simulation.

The effective temperature T_{eff} and the effective volume per unit rapidity V_{eff} are obtained by solving the coupled system of equations:

$$E = \epsilon(T_{eff})V_{eff},$$

$$S = s(T_{eff})V_{eff},$$

where the functions $\epsilon(T)$ and $s(T)$ correspond to the equation of state used in the hydrodynamic calculation.

METHOD

The charged multiplicity density per unit pseudorapidity $dN_{ch}/d\eta$ and the average transverse momentum $\langle pT \rangle$ of charged particles measured at the end of the evolution.

For each of the three equations of state, the simulation has been run for 25 values of $dN_{ch}/d\eta$, grouped in 5 sets of 5 points. Each set of 5 points is meant to simulate multiplicity fluctuations fixed collision energy, and will be used to infer the speed of sound.

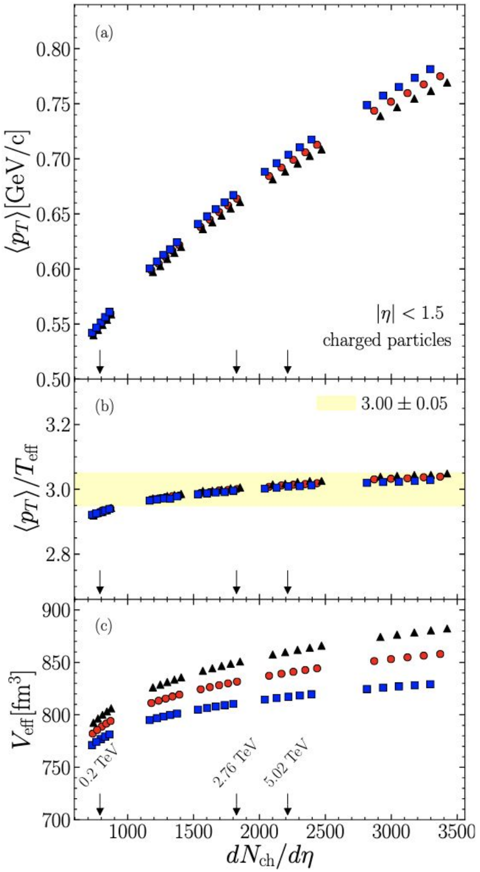
The 5 sets correspond to different collision energies. They span approximately the range $0.2 < \sqrt{s_{NN}} < 15$ TeV, from the top RHIC energy up to three times the current LHC energy.

Results

For a given equation of state, $\langle p_T \rangle$ increases as a function of $dN_{ch}/d\eta$. This increase is driven by an increase in the effective temperature T_{eff} .

The ratio $\langle p_T \rangle/T_{eff}$ is remarkably constant. It does not depend on the equation of state, and increases very mildly as a function of colliding energy.

Consistent with the choice made by CMS except for the first set of points, corresponding to the top RHIC energy, for which we assume a slightly smaller value $\langle p_T \rangle/T_{eff}$ = 2.93 ± 0.05.

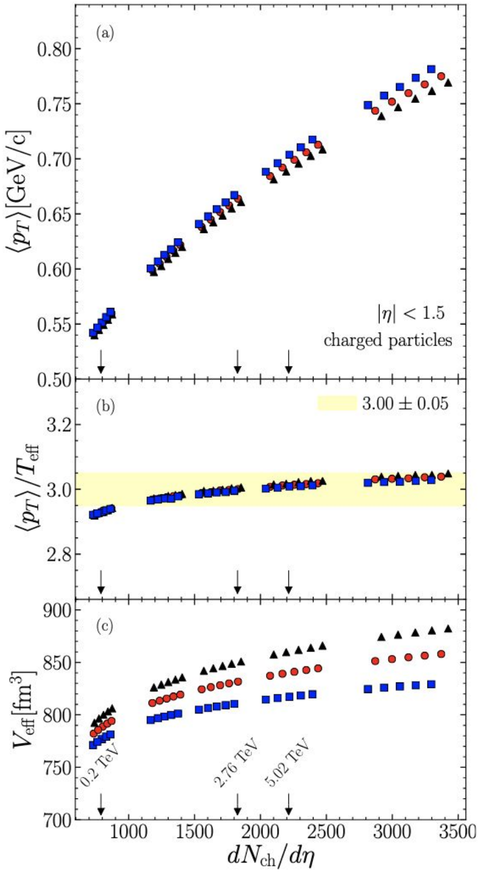


Results

V_{eff} depends somewhat on the equation of state.

For a given equation of state, the effective volume increases somewhat with the particle multiplicity,but this is a very mild increase.

In previous study, ratio $\langle p_T \rangle / T_{eff}$ was found to be insensitive to shear and bulk viscosity, and to collision centrality. So even though the initial density profile in our calculation was inferred from LHC data, the value of $\langle p_T \rangle / T_{eff}$ is robust with respect to the choice made.

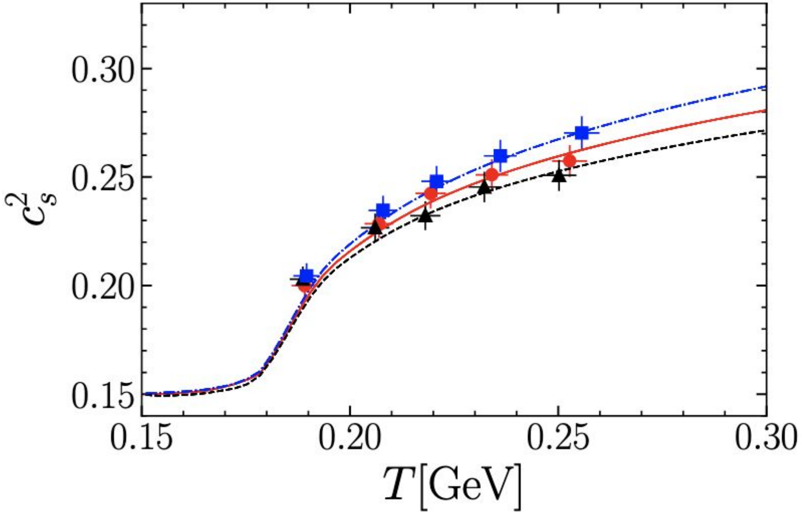


Results

Compute c^2_s by fitting each set of 5 points in Fig. using $\ln\langle pT \rangle = c^2_s \ln N_{ch}$ + constant.

The corresponding effective temperature is then evaluated from $\langle p_T \rangle$ using the proportionality relations above.

The values of $c^2_s(T)$ extracted from Eq. agree with the corresponding equations of state, within statistical error bars, for a broad range of temperatures.



Results

Note that if $\langle p_T \rangle / T_{eff}$ and V_{eff} were strictly independent of multiplicity in Fig., the agreement between the symbols and the curves in would be trivial, as it simply follows from the thermodynamic definition of c_s. But both quantities increase slightly with multiplicity,

Differences between idealized setup and an actual experiment.

Centrality determination

First, our calculation is carried out at zero impact parameter, $b = 0$. But b is not measured, and one cannot select events with $b = 0$.

Specific observable is used as a centrality classifier, for instance the transverse energy in a forward calorimeter. In each centrality bin, one then measures the average multiplicity and transverse momentum. The limit $b = 0$ is only approached asymptotically for the largest multiplicities.

Value of c^2_s depends significantly on which observable is used as a centrality classified.

Self-correlations

The dynamical fluctuation σ_{dyn} is obtained after subtracting this contribution:

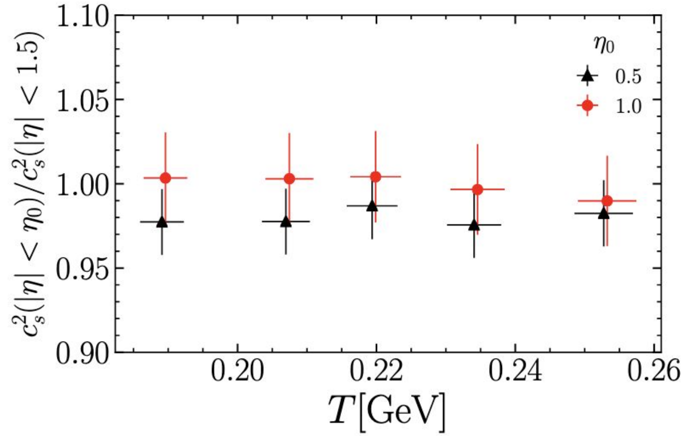
$$\sigma^2_{dyn} = \sigma^2_{Nch} - \langle N_{ch} \rangle.$$

The increase of $\langle p_T \rangle$ is driven by dynamical fluctuations only. Therefore, the derivative of $\langle p_T \rangle$ with respect to N_{ch} is decreased by a factor $\sigma_{dyn}/\sigma_{Nch}$ as the result of statistical fluctuations. As a consequence, the value of c_s extracted from Eq. is too low , and must be corrected accordingly:

$$c_s^2(T_{\text{eff}}) = \left(1 - \frac{\langle N_{ch} \rangle}{\sigma^2_{N_{ch}}}\right)^{-1/2} \frac{d \ln \langle p_T \rangle}{d \ln N_{ch}}.$$

Kinematic cuts

The next difficulties are associated with the acceptance of the analysis detector. Ideally, one should measure the average transverse momentum of all particles within a rapidity window, as the idea is to trace the total energy and entropy of the fluid, however, one only measures the momenta of charged particles above a threshold.



Local density fluctuations

- Real heavy-ion collisions have event-by-event energy density fluctuations
- These fluctuations generate anisotropic flow
- Correlation $\langle p_T \rangle$~T_{eff} remains strong even with fluctuations. Ratio $\langle p_T \rangle / T_{eff}$ largely unchanged.
- Fluctuations may modify the effective volume
- This could bias the extraction of c_s^2
- Spatial distribution of fluctuations is till worked on.

Conclusion

Conclusion of the work is that the determination of the speed of sound from experimental data seems robust and, perhaps surprisingly, precise, within a hydrodynamic description with smooth initial conditions.

Serious caveat comes from the locality of initial fluctuations, which has a number of effects.